
Supplement: Near-Parametric Learning of Autoregressive Trajectories

Pahan Dewasurendra
Johns Hopkins University

A The Conditional ρ Specialization

We give the reduction from the general results of Baraud and Birgé (2018) to the unknown-design conditional ρ theorem in the main paper. This also verifies that the unknown prompt marginal is absent from the algorithm.

Let $\mathcal{X}$ be a measurable prompt space, $\mathcal{Z}$ a finite trajectory space, and λ counting measure on $\mathcal{Z}$. Let μ be the unknown prompt marginal. A conditional density q is a measurable nonnegative function on $\mathcal{X} \times \mathcal{Z}$ satisfying $\sum_z q(z | x) = 1$ for every x . It induces a joint probability

$$Q_q(dx, dz) = \mu(dx)q(z | x).$$

The true joint probability has the disintegration $P(dx, dz) = \mu(dx)P_x(dz)$.

Lemma 1 (Conditional Hellinger identity). *For every conditional density q ,*

$$h^2(P, Q_q) = \int_{\mathcal{X}} h^2(P_x, q(\cdot | x)) d\mu(x).$$

Proof. Both probabilities are dominated by $\mu \otimes \lambda$. Expanding the definition and applying Tonelli gives

$$\begin{aligned} h^2(P, Q_q) &= \frac{1}{2} \int \sum_z \left(\sqrt{P_x(z)} - \sqrt{q(z | x)} \right)^2 d\mu(x) \\ &= \int h^2(P_x, q(\cdot | x)) d\mu(x). \end{aligned}$$

□

For n independent observations, Baraud and Birgé (2018) use the Hellinger-type distance that sums the n marginal squared distances. Thus their product-space loss is $nh^2(P, Q_q)$. Their Proposition 7 bounds the ρ -dimension of a VC-subgraph density class by

$$D_n \leq C_0(V \wedge n) [1 + \log_+(n/V)]. \quad (1)$$

Their high-probability single-model inequality, stated in Equation (20) and normalized for i.i.d. density estimation in their Equation (29), then gives

$$h^2(P, Q_{\hat{q}}) \leq C_1 \left[\inf_{q \in \mathcal{Q}} h^2(P, Q_q) + \frac{D_n + \xi}{n} \right] \quad (2)$$

with probability at least $1 - e^{-\xi}$. Their Section 8 proves that the same estimator applies to conditional densities without knowing μ .

To see this directly, every candidate joint density relative to $\mu \otimes \lambda$ is $q(z | x)$. For two candidates,

$$\frac{dQ_{q'}}{dQ_q}(x, z) = \frac{q'(z | x)}{q(z | x)}.$$

The prompt density cancels. Taking $\psi(r) = (r-1)/(r+1)$, the empirical comparison statistic is exactly

$$T_n(q, q') = \sum_{i=1}^n \psi \left(\sqrt{\frac{q'(Z_i | X_i)}{q(Z_i | X_i)}} \right).$$

It can be evaluated from the sample and the candidate conditional densities alone.

Proof of the unknown-design conditional ρ theorem.

Apply (2) and (1), take $\xi = \log(1/\delta)$, and use Lemma 1. Proposition 8 of Baraud and Birgé (2018) preserves the VC-subgraph index when passing from one-observation densities to their i.i.d. product representation. Their Theorem 3 permits a universally separable uncountable density class by reducing it to a fixed countable pointwise-dense subclass. This proves the displayed oracle inequality.

We finally verify the last proper-selection statement. Enumerate the countable subclass as $(q_j)_{j \geq 1}$. Every $T_n(q_j, q_k)$ is measurable, and so are the countable supremum over k and infimum over j . The approximate minimizer set in Definition 3 of Baraud and Birgé (2018) is nonempty. Selecting its first element in the enumeration is measurable and is itself a ρ -estimator. It lies in the candidate subclass rather than only its Hellinger closure. □

Lemma 2 (Solving the fixed-point rate). *Let $V \geq 1$, $\epsilon \in (0, 1)$, and*

$$n \geq \frac{C}{\epsilon} [V \log(C/\epsilon) + \log(1/\delta)]$$

for a sufficiently large universal C . Then

$$\frac{(V \wedge n)[1 + \log_+(n/V)] + \log(1/\delta)}{n} \leq c\epsilon$$

for any prescribed universal $c > 0$ after increasing C .

Proof. For sufficiently large C , the assumed inequality implies $n > V$. Put $r = n/V > 1$. Dropping the non-negative confidence term from the sample-size lower bound gives $r \geq C\epsilon^{-1} \log(C/\epsilon)$. Since $(1 + \log r)/r$ is decreasing for $r \geq 1$, substitution at this lower endpoint bounds it by $c\epsilon$ after increasing C . The confidence term is controlled directly by the second summand in the sample-size lower bound. $\square$

B Algebraic Trajectory Complexity

We record two variants of the trajectory-closure calculation.

Lemma 3 (Pairwise ratio subgraphs). *Under the complexity- (p, Δ) assumption, if every transition probability is strictly positive, the class*

$$\left\{ (x, z, r) : \frac{q_\theta(z | x)}{q_{\theta'}(z | x)} > r \right\}, \quad \theta, \theta' \in \Theta,$$

has VC dimension less than

$$4p \log_2(16eM\Delta).$$

Proof. Write $q_\theta = N_\theta/D_\theta$, where both trajectory polynomials have degree at most $M\Delta$. The ratio test is

$$N_\theta D_{\theta'} - r D_\theta N_{\theta'} > 0.$$

This is one polynomial inequality of degree at most $2M\Delta$ in the $2p$ parameters (θ, θ') . Applying Warren’s sign-pattern bound exactly as in the proof of the trajectory-closure theorem gives the stated bound. $\square$

This lemma explains directly why pairwise trajectory comparisons remain simple. The density-subgraph bound in the main paper is sharper and is the exact hypothesis used by Proposition 7 of Baraud and Birgé (2018).

Lemma 4 (Observed bounded stopping). *Suppose generation stops according to any rule whose decision at time t is a deterministic function of the observed prefix and whose stopping time is at most M . If the stopped path includes its length, the stopped-trajectory density class obeys the same pseudo-dimension bound as the trajectory-closure theorem.*

Proof. For a fixed stopped observation $(m, z_{1:m})$, the event that the stopping rule accepts this length is determined by $(x, z_{1:m})$ and introduces no parameter inequality. Its density is the product of the first $m \leq M$ transition probabilities, possibly multiplied by a fixed indicator. The proof of the trajectory-closure theorem applies with degree at most $m\Delta$. $\square$

B.1 Binary logistic delay line

For a fixed path, let $Y_t = \text{tail}_d(xz_{<t})$ and set $u^y = \prod_j u_j^{y_j}$. The complete density has the explicit form

$$q_u(z | x) = \frac{\prod_{t: z_t=1} u^{Y_t}}{\prod_{t=1}^M (1 + u^{Y_t})}. \quad (3)$$

The numerator degree is at most Md , and the denominator degree is at most Md . No common denominator across unobserved paths is used.

The positive rational vectors $u \in \mathbb{Q}_{>0}^d$ are pointwise dense in $(0, \infty)^d$. For every fixed path, (3) is continuous in u . Thus the universal-separability condition and the proper-selection step in the unknown-design conditional ρ theorem both hold.

B.2 Multiclass softmax

Let token zero be the reference class. For $a \in \{1, \dots, K-1\}$ and binary feature $y \in \{0, 1\}^d$, define $u_a^y = \prod_{j=1}^d u_{a,j}^{y_j}$. The conditional probabilities are

$$g_u(a | y) = \frac{u_a^y}{1 + \sum_{b=1}^{K-1} u_b^y},$$

$$g_u(0 | y) = \frac{1}{1 + \sum_{b=1}^{K-1} u_b^y}.$$

Each numerator and denominator has degree at most d in $p = (K-1)d$ positive parameters. The trajectory-closure theorem therefore gives

$$V < 2(K-1)d \log_2(8eMd).$$

The bound changes only through the free parameter count when the state feature is any fixed binary map.

C Loss Conversions

Lemma 5 (Coarsening and final-token loss). *Let P, Q be probabilities on $\mathcal{A}^M$ and let ϕ be any measurable statistic. Then*

$$h^2(P \circ \phi^{-1}, Q \circ \phi^{-1}) \leq h^2(P, Q).$$

For binary final-token probabilities p and q ,

$$(p - q)^2 \leq 2h^2(\text{Bern}(p), \text{Bern}(q)).$$

Also $\text{TV}(P, Q) \leq \sqrt{2}h(P, Q)$.

Proof. The first claim is the data-processing inequality for Hellinger distance. It also follows from Cauchy–Schwarz after grouping the summands over each fiber of ϕ . The remaining claims are the standard inequalities $\text{TV}^2 \leq 2h^2$ and $\text{TV} \leq \sqrt{2}h$ under the normalization used here. $\square$

Applying Lemma 5 pointwise in X proves the final-token claim in the near-parametric stochastic logistic CoT theorem. For trajectory sampling,

$$\begin{aligned}\mathbb{E}_X \text{TV}(P_X, Q_X) &\leq \sqrt{2} \mathbb{E}_X h(P_X, Q_X) \\ &\leq \sqrt{2 \mathbb{E}_X h^2(P_X, Q_X)}.\end{aligned}$$

Thus integrated squared Hellinger error $\epsilon^2/2$ implies expected trajectory total variation at most ϵ .

D Minimax Lower Bound

We prove the minimax parameter lower-bound theorem in full. The argument applies to every output predictor, whether or not it corresponds to a logistic generator.

Proof of the minimax parameter lower bound. First assume $d \geq 8$. Let the prompt space contain $x_1, \dots, x_d$ with $\text{tail}_d(x_j) = e_j$, and let X be uniform on these prompts. By the Varshamov–Gilbert bound, there is $\Omega \subseteq \{-1, +1\}^d$ with

$$\log |\Omega| \geq c_1 d, \quad d_H(\omega, \omega') \geq d/4$$

for distinct $\omega, \omega' \in \Omega$.

Fix $\alpha \in (0, 1/4]$ and let

$$\gamma = \log \frac{1/2 + \alpha}{1/2 - \alpha}.$$

For $\omega \in \Omega$, set $w_j^\omega = \gamma \omega_j$. At horizon one,

$$f_\omega(x_j) = \sigma(w_j^\omega) = \frac{1}{2} + \alpha \omega_j.$$

Consequently,

$$\|f_\omega - f_{\omega'}\|_{L_2(\mu)}^2 = \frac{4\alpha^2}{d} d_H(\omega, \omega') \geq \alpha^2. \quad (4)$$

Let P_ω be the law of one pair (X, Z_1) under this target. For Bernoulli parameters in $[1/4, 3/4]$,

$$\text{KL}(\text{Bern}(p) \parallel \text{Bern}(q)) \leq \frac{(p-q)^2}{q(1-q)} \leq 22\alpha^2$$

when $p, q \in \{1/2 - \alpha, 1/2 + \alpha\}$. Averaging over the prompt gives

$$\text{KL}(P_\omega \parallel P_{\omega'}) \leq 22\alpha^2. \quad (5)$$

Take $\alpha^2 = 16\epsilon$, with $\epsilon \leq 1/256$. If an estimator $\hat{f}$ has squared $L_2(\mu)$ error at most ϵ , then its nearest element of $\{f_\omega : \omega \in \Omega\}$ is the true f_ω . Indeed, the distance to the truth is at most $\sqrt{\epsilon}$, while (4) gives pairwise distance at least $4\sqrt{\epsilon}$. Fano’s inequality and (5) show that this decoding succeeds with probability at most $2/3$ whenever $n \leq c_2 d/\epsilon$. Hence no learner can guarantee error at most ϵ with success probability $3/4$ in this range.

For $d < 8$, the same order up to a universal constant follows from the two hypotheses at one coordinate below. It remains to obtain the confidence dependence. Use one prompt and two targets with probabilities $1/2 \pm \alpha$, now taking $\alpha = 2\sqrt{\epsilon}$. An estimate with squared error at most ϵ identifies the sign. The sum of the two testing errors is at least

$$\begin{aligned}\frac{1}{2} \exp[-n \text{KL}(\text{Bern}(1/2 + \alpha) \parallel \text{Bern}(1/2 - \alpha))] \\ \geq \frac{1}{2} e^{-Cn\epsilon}.\end{aligned}$$

by the Bretagnolle–Huber inequality. Making both errors at most δ requires $n \geq c_3 \log(1/\delta)/\epsilon$. Taking the larger of the dimension and confidence lower bounds, then changing the universal constant, proves the stated lower bound. $\square$

E Sparse Model Selection

For $S \subseteq [d]$, let $\mathcal{Q}_S$ be the logistic trajectory family with $w_j = 0$ for $j \notin S$. Put $s = |S|$ and

$$V_s = \lceil 2(s \vee 1) \log_2(8eM(s \vee 1)) \rceil.$$

The harmless use of $s \vee 1$ includes the all-zero model.

Choose model weights

$$\Delta(S) = \log \binom{d}{|S|} + 2 \log(|S| + 1) + \log(\pi^2/6). \quad (6)$$

They satisfy

$$\sum_{S \subseteq [d]} e^{-\Delta(S)} = \frac{6}{\pi^2} \sum_{s=0}^d \frac{1}{(s+1)^2} \leq 1.$$

Theorem 6 (Sparse trajectory oracle). *There is one penalized conditional ρ -estimator such that, with probability at least $1 - \delta$,*

$$\begin{aligned}\mathbb{E}_X h^2(P_X, \hat{q}_X) \leq C \inf_{S \subseteq [d]} \left\{ \inf_{q \in \mathcal{Q}_S} \mathbb{E}_X h^2(P_X, q_X) \right. \\ \left. + \frac{V_s[1 + \log_+(n/V_s)]}{n} \right. \\ \left. + \frac{\Delta(S) + \log(1/\delta)}{n} \right\}.\end{aligned}$$

Proof. Each fixed-support class is universally separable and has pseudo-dimension at most V_s by the trajectory-closure theorem. Apply the penalized model-selection theorem of Baraud and Birgé (2018, Theorem 2) with the Kraft weights (6). Its model complexity is bounded by Proposition 7 as in (1). Divide the product-space Hellinger loss by n and apply Lemma 1. $\square$

If the truth has support size s , use $\Delta(S) \leq s \log(ed/s) + 2 \log(s+1) + C$. Lemma 2 then yields the sparse rate displayed in the main paper up to secondary logarithms.

F Misspecification and Contamination

The approximation term in the unknown-design conditional ρ theorem already proves the misspecified claim. We add details for Huber contamination because the contaminating distribution may alter the prompt marginal.

Let P be the clean joint law of a prompt and trajectory, and let

$$R = (1 - \eta)P + \eta H$$

be the observed law. Denote their prompt marginals by μ_P and μ_R . Assume the clean conditional density is $q^* \in \mathcal{Q}$. The candidate joint law used by the conditional estimator is

$$Q_R^*(dx, dz) = \mu_R(dx)q^*(z | x).$$

Since $\text{TV}(P, R) \leq \eta$ and marginalization contracts total variation,

$$\begin{aligned} h^2(P, R) &\leq \eta, & h^2(\mu_P, \mu_R) &\leq \eta, \\ h^2(P, Q_R^*) &= h^2(\mu_P q^*, \mu_R q^*) = h^2(\mu_P, \mu_R) \leq \eta. \end{aligned}$$

The squared triangle inequality gives $h^2(R, Q_R^*) \leq 4\eta$. The unknown-design conditional ρ theorem applied to the observed law therefore gives

$$h^2(R, \mu_R \hat{q}) \leq C(\eta + \mathfrak{C}_n)$$

with high probability, where $\mathfrak{C}_n$ is the estimation term. Finally,

$$\begin{aligned} h(P, \mu_P \hat{q}) &\leq h(P, R) + h(R, \mu_R \hat{q}) \\ &\quad + h(\mu_R \hat{q}, \mu_P \hat{q}), \end{aligned}$$

and the last term equals $h(\mu_R, \mu_P)$. Squaring proves

$$\mathbb{E}_{X \sim \mu_P} h^2(P_X, \hat{q}_X) \leq C'(\eta + \mathfrak{C}_n).$$

This is the contamination statement in the main paper.

G Why the e2e Calculation Differs

For one observed CoT path, Equation (3) has one denominator product of degree at most Md . Under e2e supervision, the observed density is instead

$$q_u^{\text{e2e}}(1 | x) = \sum_{z_{1:M}: z_M=1} q_u(z_{1:M} | x).$$

The summands have path-dependent denominators. The proof of Doron-Arad et al. (2026, Lemma 12) clears them using

$$D(u) = \prod_{y \in \{0,1\}^d} (1 + u^y)^M,$$

whose degree is $Md2^{d-1}$. The Goldberg–Jerrum bound then becomes

$$O(d \log(Md2^d)) = O(d^2 + d \log M).$$

Our observed-path closure theorem does not change this sum-before-threshold operation. Any improvement for e2e supervision requires additional structure beyond the present algebraic argument.

References

- Baraud, Y. and Birgé, L. (2018). Rho-estimators revisited: General theory and applications. *The Annals of Statistics*, 46(6B):3767–3804.
- Doron-Arad, I., Mehal, I., and Mossel, E. (2026). Stochastic autoregressive learning. *arXiv preprint arXiv:2608.07224*.